Asynchronous Physical Exchanges

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4. Conceptual Technical Architecture

4.1 Separation of responsibilities of complementary layers

The process relies on an explicit separation between two complementary technical layers, each having a distinct and non-substitutable role.

Layer A corresponds to the monetary settlement layer. It relies on a truly decentralized public blockchain, resistant to censorship and based on strict consensus rules, such as Bitcoin. This layer is used exclusively for irreversible payments and the economic commitments associated with the different stages of the process.

Layer B corresponds to a distributed cryptographic infrastructure responsible for the coordination of physical states. It is linked to physical lockers and allows the recording, broadcasting and comparison of observable physical characteristics associated with exchanged objects. This layer manages no transfer of monetary value and does not interpret the nature of exchanges; it is limited to state management and the attestation of observable material facts.

This separation aims to avoid any confusion between cryptographic proof and material reality. Layer A guarantees economic irreversibility, while Layer B provides a coordination framework allowing physical events to be linked to cryptographic commitments without introducing an arbitral third party.

4.2 Settlement layer: economic commitments and irreversible payments

Layer A of settlement ensures all economic commitments of the process through mechanisms already existing within the Bitcoin protocol, notably conditional transactions and time locks.

Logistics commitment:

When the object is taken in charge by logistics, the operator produces an explicit cryptographic commitment on Layer A. This commitment takes the form of a conditional transaction locked by a time constraint.

Concretely, a predefined amount is committed such that:

• if no success condition is fulfilled before the deadline

• then the transaction automatically becomes executable in favor of the buyer.

This mechanism requires no later decision or human intervention: the time expiration itself constitutes a sufficient condition.

Case of non-delivery or delay:

If logistics does not deposit the object in a reception locker within the allotted time, no additional action is required.

The simple passage of time releases the time lock, making the transaction executable according to the predefined rules. The reimbursement of the buyer is then triggered automatically by Layer A.

Case of physical non-conformity:

Upon arrival, the reception locker performs a new acquisition of observable physical characteristics of the object. These data are compared to the characteristics recorded during the initial deposit and explicitly validated by the buyer at that moment.

It is fundamental to emphasize that:

• the locker does not decide the conformity of the object

• it interprets no rule

• it resolves no dispute

It strictly limits itself to a mechanical comparison between two sets of data:

• the physical data accepted by the buyer during the initial deposit

• the physical data observed at reception

If the sets do not correspond, the object is considered non-conforming according to the criteria already validated by the buyer. The locker merely attests this divergence.

In this case, no direct signal is sent to the settlement blockchain. Since the object is refused, no success condition is fulfilled, and the time mechanism initiated by logistics simply continues until expiration. At the end of the delay, the conditional transaction becomes executable, triggering the automatic reimbursement of the buyer.

Case of success and cancellation of the time lock:

In the case of conformity, the object is accepted by the reception locker based on the data validated ex ante by the buyer. This acceptance results in the revelation of a cryptographic secret previously committed, allowing the cryptographic condition that invalidates the penalizing branch of the conditional transaction engaged by logistics to be satisfied.

This mechanism relies on classical hashlock / timelock constructions:

• the revelation of the secret prevents execution of the reimbursement

• the economic commitment of logistics is then considered fulfilled

This is not a decision oracle, but simply the lifting of a cryptographic condition defined from the beginning.

Absence of trusted third party:

At no moment does Layer B impose a decision on Layer A. It merely attests observable material facts and reveals, or not, cryptographic elements defined in advance. Economic rules are entirely defined at the initial commitment, and their execution relies exclusively on the protocol of the settlement blockchain.

Thus the process does not rely on a trusted third party, but on:

• irreversible economic commitments

• objective temporal constraints

4.3 Distributed physical state infrastructure layer

Layer B constitutes a distributed cryptographic infrastructure responsible for representing, recording and comparing observable physical states associated with material objects. It does not constitute a monetary blockchain but a registry of verifiable physical states whose function is to coordinate interactions between the material world and the cryptographic system.

Its role is not to issue judgments nor to take arbitrary decisions, but only to observe, record and compare measured observations.

The Layer B registry is of the append-only type:

data can be added to it but never modified or deleted.

This property guarantees:

• the historical integrity of records.

• the impossibility of retroactively falsifying a measurement.

• the public verifiability of past states.

Each recorded state therefore becomes an immutable timestamped proof.

When an exchange is initiated, the system generates a unique cryptographic identifier. This identifier is derived from parameters such as:

• the public keys of the parties.

• a random uniqueness factor.

• a timestamp.

This identifier reveals no personal information but makes it possible to associate with certainty:

• a seller.

• a buyer.

• an object.

• an exchange session.

It constitutes the common reference for the entire process.

Layer B relies on a distributed network of independent nodes that may be operated by different actors such as:

• locker manufacturers.

• logistics operators.

• technical institutions.

• independent operators / individuals.

Each node holds a copy of the registry and participates in the validation of new states.

The role of a node is strictly defined:

• verify the cryptographic validity of entries.

• verify temporal consistency.

• record new observations.

• refuse any modification of a past state.

No node possesses privileged authority.

Each user of the system possesses a personal cryptographic key allowing them to interact with the infrastructure regardless of their role. The same individual may therefore act successively as buyer or seller without changing cryptographic identity.

Logistics operators possess distinct professional keys used only to sign their transport commitments.

These keys may be stored on secure hardware supports such as:

• cryptographic cards.

• secure modules.

This approach avoids any dependence on a personal device or on centralized infrastructure.

During the initial deposit of an object, the locker automatically performs a multi-source acquisition of observable physical characteristics. Depending on the available equipment, these measurements may include:

• mass.

• external dimensions.

• volume.

• optical imaging.

• internal imaging (tomography or X-rays).

• material signatures.

• density distribution.

These data do not constitute a subjective visual description but a set of objective quantitative measurements.

The collected measurements are transformed into a standardized digital representation and then condensed into a unique cryptographic fingerprint.

This fingerprint constitutes a physical identifier mathematically derived from the measured properties of the object. It does not allow the object to be reconstructed but makes it possible to verify whether two observations correspond to the same physical state.

The data measured during the deposit are transmitted to the buyer via Layer B. The buyer examines this information and decides whether to accept or reject it.

Acceptance means that the buyer recognizes that the recorded physical fingerprint corresponds to the expected object. This validation becomes the reference for comparison for all future verifications.

From this moment onward, Layer B never interprets the data: it only compares subsequent measurements with this accepted reference.

During delivery, the destination locker performs a new complete acquisition of physical characteristics according to the same process used during the initial deposit.

A mathematical comparison is then performed between:

• the fingerprint recorded at deposit.

• the fingerprint measured at reception.

Only two outcomes are possible:

• match: the physical state is identical.

• mismatch: the state has been modified.

The system produces no judgment: it only observes equality or inequality of the fingerprints.

The locker is not a trusted third party, an arbitrator, or a decision authority.

It performs only three functions:

• measure.

• compare.

• attest the result.

The logical outcome results exclusively from the mathematical comparison between the initial fingerprint validated by the buyer and the fingerprint measured at reception.

The locker does not interpret the results: it mechanically applies the predefined rule.

Layer B does not claim to demonstrate the internal functioning of an object nor its functional state. It only makes it possible to attest whether a received physical object corresponds or does not correspond to the physical state observed during the initial deposit.

It therefore constitutes a risk-reduction mechanism, not an absolute proof.

4.4 Distributed validation of Layer B states

The validity of a state recorded in Layer B does not depend on any central authority nor on a single actor. It relies on a distributed validation mechanism ensured by the network of independent nodes maintaining the registry.

When a new event is produced — session creation, measured deposit, buyer validation, logistics pickup, reception measurement — this event is broadcast to the network in the form of a cryptographically signed entry. Each node then independently verifies:

• the validity of signatures.

• the temporal consistency of the event.

• conformity to the expected format.

• compatibility with the previously recorded state.

An event is accepted into the registry only if it satisfies all of these conditions. Otherwise, it is locally rejected by the nodes.

Nodes never validate physical reality itself. They do not verify the object, perform no measurements, and do not judge the results. Their role consists exclusively of validating the cryptographic and logical consistency of the statements produced by measurement devices. In other words, they validate proofs of observation, not the observation itself.

The consensus of Layer B therefore concerns only the order and authenticity of recorded states, not the interpretation of material reality. This distinction ensures that the infrastructure remains a distributed verification system and not a decision-making mechanism.

When a valid event is accepted by the network, it is added to the append-only registry and becomes an immutable reference. The more a state is replicated and confirmed by nodes, the stronger its finality becomes. After a certain number of confirmations, a state is considered stable and can no longer be contested without invalidating the entire subsequent history.

This model makes it possible to obtain an essential property:

no isolated actor — neither a node, nor an operator, nor a locker manufacturer — can modify, delete or falsify a recorded state.

Trust therefore does not rely on a participant, but on the distributed structure of the system itself.

5. System resilience and security limits

The security of Layer B relies on a combination of cryptographic guarantees and physical constraints. Unlike purely digital distributed systems, the initial state recorded in Layer B is not an abstract data generated by computation, but the result of a measurable material observation performed by a physical device.

Each state record corresponds to a real event that occurred in the physical world. This property introduces a fundamental constraint: the falsification of a valid state requires an effective material interaction with the infrastructure, and cannot be obtained solely through software manipulation of the network.

5.1 Resistance to identity attacks (Sybil)

In a classical distributed system, an attacker may attempt to take control of the network by multiplying identities. This type of attack aims to influence consensus or the recording of data.

In Layer B, the creation of additional network identities does not make it possible to fabricate false physical states, because:

• measurements are produced by real hardware devices.

• each observation is locally signed by a cryptographic module.

• recorded states are immutable and publicly verifiable.

Thus, multiplying nodes does not make it possible to generate false physical proofs. At most it can disturb the propagation or availability of data, but it cannot retroactively alter a state already validated.

5.2 Physical anchoring of proofs

The fundamental property ensuring the robustness of the system is the physical anchoring of states.

A state is accepted by the network only if it corresponds to a measured and signed observation. The very existence of this state therefore implies that a material event has actually occurred.

This constraint introduces an irreducible real cost for any attempt at falsification, since an attacker would have to compromise:

• a measurement device.

• a hardware cryptographic module.

• or the physical infrastructure itself.

Security therefore does not rely only on a majority of honest nodes, but also on the practical difficulty of reproducing or falsifying a measurable physical event.

5.3 Inherent limits of the model

Layer B does not claim to prevent all attacks. It does not eliminate:

• attempts of network censorship.

• propagation delays.

• attacks on hardware infrastructure.

• physical or technical failures.

It also does not guarantee the internal functioning of an object, but only the measurable correspondence between two observed physical states at distinct moments.

The system therefore constitutes a mechanism of risk reduction, not a mechanism of absolute certainty.

5.4 Separation between physical truth and network consensus

The role of the network is not to determine physical reality, but to record in an immutable way the observations produced.

In other words:

• material reality is established by measurement.

• the network merely attests that this measurement has taken place.

This separation prevents distributed consensus from becoming an arbitrary authority over the state of an object. Nodes do not decide the validity of a physical state; they only verify that this state has been correctly produced and signed.

5.5 Emergent property: hybrid security

The resulting architecture possesses a particular property: it combines two distinct and complementary sources of security:

• mathematical security (cryptography, signatures, immutability)

• physical security (material cost, real constraints, irreversibility of events)

This hybridization makes purely digital attacks insufficient to compromise the system as a whole.

6. Scope, implications and positioning of the model

6.1 Decentralized nature of the process

The proposed model relies on a distributed architecture in which no single entity holds a global decision-making power.

Layer A relies on a decentralized monetary protocol.

Layer B relies on a distributed registry whose states are validated collectively by independent nodes.

Physical devices measure, nodes record, actors sign.

No central authority interprets, arbitrates or validates on behalf of others.

The process therefore does not rely on trust in an institution, but on:

• cryptographic verifiability.

• immutability of records.

• measurable physical constraint.

• strict separation of responsibilities.

6.2 Conceptual scope

This document describes a theoretical model aimed at structuring asynchronous physical exchanges without dependence on a central arbitral third party.

It proposes an architecture in which:

• value is settled on a decentralized monetary infrastructure.

• physical state is attested through measured observations.

• the coherence of events is ensured by a distributed registry.

This is not a product nor an existing implementation, but a conceptual framework open to discussion, experimentation and improvement.

6.3 Potential domains of application

The model could apply to any situation involving:

• an exchange deferred in time.

• an absence of prior trust.

• a need for objective verifiability of a material state.

This potentially includes:

• peer-to-peer exchanges.

• logistics with conditional liability.

• transport of sensitive objects.

• distributed supply chains.

These applications do not constitute an imposed objective but possible illustrations.

6.4 Practical limits and implementation constraints

Such a system would require:

• the deployment of specialized hardware infrastructures.

• the integration of reliable measurement devices.

• technical standardization.

• coordination between operators.

These constraints relate to industrial and economic considerations, not to the logical coherence of the model.

6.5 Assumed structural limits

The model does not claim to:

• eliminate all risk.

• prevent all fraud.

• nor replace existing legal frameworks.

It does not demonstrate the intrinsic nature of an object nor its internal functioning.

It only attests the measurable correspondence between two observed physical states.

It therefore constitutes a mechanism of uncertainty reduction, not an absolute guarantee.

6.6 Intellectual positioning

This document should be understood as:

• an architectural exploration.

• a technical proposal.

• an open framework for reflection.

It describes a structural possibility.

The problem of exchanges of physical goods associated with cryptographic mechanisms has already been the subject of various works and proposals. However, to my knowledge, although I have not had the time nor the opportunity to examine all existing works, the approaches I have seen do not propose the architectural combination presented here.